International Journal of Psychophysiology 95 (2015) 167–174

Contents lists available at ScienceDirect

International Journal of Psychophysiology

journal homepage: www.elsevier.com/locate/ijpsycho

Speaker's voice as a memory cue

Sandra Campeanu [a,b], Fergus I.M. Craik [a,b], Claude Alain [a,b,c,*]

[a] Rotman Research Institute, Baycrest Centre, Toronto, ON, Canada
[b] Department of Psychology, University of Toronto, ON, Canada
[c] Institute of Medical Sciences, University of Toronto, ON, Canada

ARTICLE INFO

Article history:
Received 28 January 2014
Received in revised form 14 August 2014
Accepted 19 August 2014
Available online 28 August 2014

Keywords:
Voice
Event-related potential
Memory
Context
Recollection
Attention

ABSTRACT

Speaker's voice occupies a central role as the cornerstone of auditory social interaction. Here, we review the evidence suggesting that speaker's voice constitutes an integral context cue in auditory memory. Investigation into the nature of voice representation as a memory cue is essential to understanding auditory memory and the neural correlates which underlie it. Evidence from behavioral and electrophysiological studies suggest that while specific voice reinstatement (i.e., same speaker) often appears to facilitate word memory even without attention to voice at study, the presence of a partial benefit of similar voices between study and test is less clear. In terms of explicit memory experiments utilizing unfamiliar voices, encoding methods appear to play a pivotal role. Voice congruency effects have been found when voice is specifically attended at study (i.e., when relatively shallow, perceptual encoding takes place). These behavioral findings coincide with neural indices of memory performance such as the parietal old/new recollection effect and the late right frontal effect. The former distinguishes between correctly identified old words and correctly identified new words, and reflects voice congruency only when voice is attended at study. Characterization of the latter likely depends upon voice memory, rather than word memory. There is also evidence to suggest that voice effects can be found in implicit memory paradigms. However, the presence of voice effects appears to depend greatly on the task employed. Using a word identification task, perceptual similarity between study and test conditions is, like for explicit memory tests, crucial. In addition, the type of noise employed appears to have a differential effect. While voice effects have been observed when white noise is used at both study and test, using multi-talker babble does not confer the same results. In terms of neuroimaging research modulations, characterization of an implicit memory effect reflective of voice congruency is currently lacking.

1. Introduction

Speaker's voice is well known to influence attention and memory. For instance, in noisy environments we more easily understand what a person is saying if the voice is familiar (Rosenblum et al., 2007). A familiar voice also helps us locate friends in a crowded room (Johnsrude et al., 2013). These examples highlight the impact of voice in guiding attentional processes, but the effect of voice congruency on memory is less clear. One major area of interest in memory research has been to identify factors that facilitate retrieval of information. In the context of auditory verbal memory, speaker's voice likely occupies a central role in modulating memory performance. Investigation into the nature of voice representation as a memory cue is essential to understanding auditory memory and the neural correlates which underlie it. Here, we review studies that have examined the effects of speaker's voice on memory performance in an effort to answer the following questions. Do people remember words better if they are repeated in the same voice? What about a similar voice? Does any such benefit of speaker's voice on memory performance depend on explicit attention to the voice? The present work aims to develop a recipe for finding voice effects in auditory memory studies, using neutral stimuli, in young adults, and to highlight future directions of inquiry — especially in an effort to understand the neural correlates that might underlie these voice effects in various situations. The investigation into voice – arguably the most ecologically important source in auditory scene analysis – as an auditory memory cue began some forty years ago.

To our knowledge, the study from Craik and Kirsner (1974) was among the first to examine the role of voice reinstatement on memory performance for spoken words. Using a continuous recognition paradigm, they found that participants were better and faster at recognizing words spoken in the same voice at test as at study. Though the effect was relatively small, the conclusion was that voice information persists over time and that context congruency (i.e., same speaker between study and test) facilitates verbal memory. The notion that context congruency facilitates memory performance is consistent with the encoding specificity principle (Tulving and Thomson, 1973). The principle states that the details of encoding dictate what is stored, which in turn dictates the cues that are effective in retrieval. In the case of word

* Corresponding author at: Rotman Research Institute, Baycrest Centre, Toronto, Ontario M6A 2E1, Canada. Tel.: +1 416 785 2500x3523.
E-mail address: calain@research.baycrest.org (C. Alain).

http://dx.doi.org/10.1016/j.ijpsycho.2014.08.988
0167-8760/

recognition, items are better remembered if the test episode is as close as possible to the study episode; for example, using the same speaker's voice between study and test facilitates word memory.

The findings from Craik and Kirsner (1974) raise several questions. First, does partial voice congruency provide a partial facilitation effect, or is full voice reinstatement required to see a memory benefit? The literature addressing this question has provided mixed results, as described later in the article (Palmeri et al., 1993 vs. Goldinger, 1996). Second, what circumstances (if any) modulate this facilitation effect? Third, does attention to voice matter, or are the beneficial effects of voice encoded automatically? Finally, is facilitation present for implicit memory as well, or only for explicit recognition memory?

In the following sections we review empirical findings and theoretical suggestions relevant to the use of voice as a memory cue in explicit and implicit auditory memory. With respect to explicit memory, we discuss word recognition and voice recognition and summarize neural correlates found to underlie the use of voice in such cases. With respect to implicit memory, we discuss voice-related findings found using indirect memory tests in the auditory domain. In both kinds of memory, we discuss the effects of attention allocation and expertise, in an effort to delineate conditions that promote beneficial voice effects. In conclusion, we summarize findings and suggest future directions for research investigating voice effects in memory.

2. Voice effects in explicit memory

As mentioned above, the study by Craik and Kirsner (1974) led the way for investigations into the role of voice in explicit memory. Pisoni (1993) conducted a review and concluded that detailed perceptual information about speaker voice, such as gender and dialect of the speaker, is encoded automatically and in parallel with the episodic memory trace of the word itself, and is therefore retained in long term memory. However, further research into the role of voice in word and voice memory provided mixed results.

Though they did find same-voice facilitation for word memory, Craik and Kirsner did not find a same-voice benefit for memory of the voice itself. Geiselman and Bellezza (1976, 1977) then suggested that gender of the speaker voice is automatically encoded. In their first study, Geiselman and Bellezza (1976) investigated the effect of adding an explicit voice recognition task to a sentence memory test. After free recall of sentences that had previously been heard, participants made a voice judgment for each sentence (male or female). Critically, some participants were told about the subsequent source test (intentional condition), while some were not (incidental condition). It was hypothesized that if speaker voice was encoded automatically, then participants in the incidental condition would demonstrate speaker voice recognition without a corresponding decrease in sentence recall. In fact, participants in both incidental and intentional conditions were able to recognize the voice attribute (i.e., gender) above chance levels, and did so without suffering a decrease in sentence recall performance. The authors suggested that this automaticity of voice information might be due to an integral incorporation of the voice into the code, rather than a mere attachment to the memory trace. In their subsequent study, Geiselman and Bellezza (1977) investigated whether such integration would mean that the connotation of the sentence and its relation to the gender of the speaker's voice, would make a difference. Participants heard sentences spoken by male or female voices. The sentences either contained gender-specific nouns or gender-neutral nouns. Results indicated that voice retention was superior when sentence stimuli were gender-neutral. Apparently voice is automatically encoded when neutral stimuli are employed. The authors suggested a voice connotation hypothesis — speaker voice interacts with the semantic interpretation of the sentences; automatic encoding that precludes such interaction occurs when the sentences are gender-neutral. These findings suggested that the gender of a voice is automatically encoded, and it followed that the lack of voice recognition effects in the Craik and Kirsner (1974) study might reflect a gender confound. That is, the different speaker condition in the Craik and Kirsner investigation always corresponded to a gender change, making it so that participants were just as likely to detect congruity as incongruity of the voice. Whether more detailed voice information is encoded or not, remained to be investigated.

Palmeri, Goldinger and Pisoni (1993) investigated this possibility in a follow-up to Craik and Kirsner's (1974) investigation. Since only two voices were used in the initial study by Craik and Kirsner – one of each gender – the study by Palmeri et al. (1993) sought to clarify a potential gender confound and to extend the investigation into multiple talker conditions. Using a continuous recognition paradigm, Palmeri et al. found that words repeated in the same voice during the second presentation as in the first were better recognized than words re-presented in different voices, irrespective of the gender of the voice. This finding showed that more detailed physical information, not only gender of the speaker, is encoded with the word. The investigators also manipulated the number of voices used in each continuous stream and found that talker variability did not affect performance. Interestingly, words were not better remembered if the voice at test was of the same gender as at study. Therefore, the study by Palmeri et al. indicated no partial benefit in word recognition for similar voices — at least when the partial benefit came from gender.

Palmeri et al. (1993) also concurrently measured voice recognition. Participants were asked to identify whether a test word was new, old with the same voice as at study, or old with a different voice than at study. Results indicated that a benefit in voice (or "source") memory occurred when words were presented in the same voice at study and at test. Interestingly, the study concluded that voice recognition was more accurate when words were re-presented in either the same voice or in different voices of different genders rather than in different voices of the same gender. There was, however, no difference between performance when the same voice was used and when different voices of different genders were used. This last finding echoed Craik and Kirsner's (1974) finding with respect to voice recognition. Speculatively, it may be that different voice trials can be just as perceptually salient as same voice trials, if the differences in voice characteristics are substantial (i.e., across gender).

A few years later, Dodson et al. (1998) reported evidence in support of this conjecture when they investigated memory for source information (voice) during a divided attention task. They found that memory for the voice (so-called specific source information) was compromised when attention was divided, while retrieval of the gender of the voice (so-called partial source information) remained intact. From this result, it appears that gender is a strong if general vocal cue that is automatically encoded with the memory trace, while attention plays a pivotal role in more specific voice recognition.

Therefore, an important factor in the Palmeri et al. (1993) investigation was allocation of participants' attention. Specifically, participants did not expressly attend to voice. The finding of a same speaker advantage but no partial same gender benefit might have been contingent on attention, with the exact voice reinstatement benefit in word recognition being fairly automatic while partial benefits associated with lesser voice congruency might depend on attention allocation. Indeed, and in contrast to the notion of automaticity of voice information, several studies have pointed to the importance of attention allocation for finding voice effects. In a series of experiments, Goldinger (1996) explored voice congruency as a function of levels of processing and type of memory investigated. In a first experiment, he devised a similarity matrix of voices based on participant responses — which classified voices on an ordinal scale based on pitch. This matrix was used to rate voice congruency between study and test in the subsequent experiments. Using surprise recognition tasks, Goldinger found that reinstating the same voice at test as at study improved word recognition. In addition, recognition was better when words were re-presented in a different voice of the same gender than when they were re-presented in a different voice of a different gender, implying that degrees of voice congruency may affect memory for words. That is, the identical voice condition yielded the best word recognition, followed by the similar voice condition (same gender), and then, lastly, the dissimilar voice condition (different gender). This finding appears to contradict Palmeri et al.'s (1993) null partial congruency finding, but one must consider that Goldinger used a different (block) study design, with specific attentional focus at study. With respect to explicit memory, Goldinger found that deeper levels of encoding led to better overall recognition but that deeper encoding tasks were associated with a smaller voice effect. Given that the shallowest level of processing drew participants' attention to voice characteristics (gender classification) while the deepest level of processing focused on syntax, it is perhaps not surprising that the latter yielded a smaller voice effect at test. In fact, the finding that attending to voice characteristics makes participants more sensitive to voice changes falls in line with the encoding specificity principle (Tulving and Thomson, 1973), and with the transfer-appropriate processing (TAP) framework put forth by Morris et al. (1977). The TAP framework suggests that congruency between encoding features and subsequent retrieval task yields better memory. That is, when visual features are encoded, for example, performance on a visual memory task will be enhanced. Appropriately, it appears that voice effects are more prominent when voice characteristics are emphasized at encoding. In the Goldinger (1996) study, the benefits of voice congruency on recognition memory lasted at least a full day but less than a week. Attentional focus was a key factor in this investigation; as Goldinger (1996) says, “it appears that the content of study words' episodic traces is influenced by the focus of attention at study” (p. 1177).

Johnson et al. (1996) similarly found that attentional focus at encoding is key to determining subsequent memory effects. These authors looked at emotional focus and found that sentence memory was at least as good when participants focused on how they felt about what was being said as when they focused on the speaker's emotions. However, source memory performance was worse when participants focused on their own emotions regarding the content of what was heard — an effect that could be attenuated if participants focused on how they felt about the speakers rather than on the contents of the sentences. Overall, it seemed that participants performed better on memory tasks that tapped the components at focus during encoding; attention to speakers bolstered source memory.

Earlier work by Fisher and Cuervo (1983) had suggested that the amount of information about speaker's voice that is generally retained is relatively small because perceptual attributes are not particularly meaningful. These authors suggested that semantic, conceptual properties are more meaningful and better encoded (also discussed in Goldinger, 1996). It is therefore crucial to underline the importance of instructions at study and of correspondence between study and type of memory test, in order to direct participants' attention appropriately in order to obtain voice effects. The voice congruency effect is likely paradigm-dependent and one might expect to find voice effects on word recognition in studies where perceptual attributes of speaker voice are emphasized and/or where processing is shallow. For example, Bradlow et al. (1999) found a same-voice facilitation effect for word memory, in that same-speaker words were better remembered than different-speaker words across lags in the continuous recognition paradigm used (which varied voice, speaking rate and amplitude — all perceptual attributes).

Although there is evidence that voice congruency facilitates recognition, there are several studies that failed to observe a same-voice benefit for word memory. In investigating the time course of neural responses associated with word and voice memory, Senkfor and Van Petten (1998) found that word recognition did not differ as a function of voice congruency. Here, participants were told to form a mental image of the word at study and make a size judgment about it, which focused participants on conceptual properties at encoding. Naveh-Benjamin and Craik (1995, 1996) also showed negative findings in this respect. Naveh-Benjamin and Craik (1995) used a study–test design that they suggest may have resulted in no same-voice advantage because of longer retention intervals (and a distractor task which was employed). Naveh-Benjamin and Craik (1996) investigated perceptual and conceptual encoding and the consequent effects on memory in aging. Considering only the young adult group for the present purposes, no same-voice benefit was found for item recognition. Like Craik and Kirsner (1974), the study by Naveh-Benjamin and Craik (1996) also reported no same-voice source recognition benefit. Though participants were instructed to consider voice at study, they did not expect a subsequent voice test. It may be that performance on a source recognition test depends greatly on explicit instructions to the participants; that is, when participants are directed to focus attention to the speaker in preparation for a source recognition test, the same-voice source recognition facilitation seems more likely — as previously found by Palmeri et al. (1993).

Schacter and Church (1992) also reported no significant same-voice benefit for recognition; they suggested that the lack of memory facilitation provided by voice congruency between study and test may reflect participants' increased reliance on conceptual rather than perceptual encoding. There is some evidence (e.g., Goldinger, 1996, above) of a difference between perceptual and conceptual information use in memory, such that perceptual encoding informs perceptual sensitivity at retrieval (and conceptual congruency works in a similar manner). Schacter and Church did, nonetheless, report two experimental manipulations that found no same-voice benefit in explicit word memory. First, they utilized pitch discrimination as an encoding task and found no voice effect in word recognition. They suggested that perceptual emphasis might need to be in relation to a particular word, rather than a non-specific acoustic judgment such as pitch discrimination, in order to see voice effects on recognition memory — a concept which supports the idea that word and voice are integrated into a memory trace. Their second manipulation involved a clarity-rating encoding task in which participants had to judge how well voices enunciated each word. Unfortunately, when investigating explicit memory in this design, a cued-recall test was employed and it followed a distractor task and an implicit memory task, making a comparison of these findings to the current review inappropriate. Church and Schacter (1994) subsequently looked at acoustic variation in implicit memory in particular, but also included a recognition task. No same-voice benefit for item recognition was observed (Experiment 1). However, one could argue that the distractor task and implicit memory task, both of which occurred between study and the recognition test, may have had a disruptive effect. Pilotti and Beyer (2002) performed a recognition memory task, which compared the processing of different types of information, including voice, in implicit and explicit memory. The word recognition task (Experiment 3) indicated no same-voice benefit between study and test. However, a familiarization phase was used before the task so that participants might become acquainted with the voices used. This likely confounded a straight comparison of voice effects.

Lastly, Dodson and Shimamura (2000) also reported no effect of voice congruency on word recognition. In this study, participants were told to imagine the speaker saying the word at encoding, and then were tested on word and source memory. Despite their being no benefit of voice congruency on word memory, participants did show a benefit of voice congruency in voice recognition. Critically, words were presented in the auditory and in the visual domain at both study and test, so that a control condition involved a test word being presented only visually. In the source recognition test, voice incongruency actually led to weaker voice recognition than the control condition (using a novel voice yielded results similar to the control condition). A follow-up study by Dodson (2007) concluded that this deterioration in voice recognition when study and test voices do not match results from participants experiencing illusory recollections based on the retrieval environment at test. However, it is important to note that these experiments utilized a visual presentation of the stimuli, thereby providing a more complex memory trace than that formed by a word in a purely auditory scene. Regardless, Roediger et al. (2004) found evidence that participants tend to form illusory recollections for voices in a purely auditory test (Experiment 4). Interestingly, this effect might not apply to word recognition, since Goh (2005) showed no benefit of novel voices over different, studied voices in a word memory task.

Taken together, voice effects have sometimes been lacking in recognition studies, indicating a susceptibility of voice effects to paradigm design. These findings challenged the idea that same-voice facilitation in memory is automatic, suggesting that some attention to the perceptual features of the bound trace (of a word and the voice saying it) as well as a lack of interference (or some minimum combination of these two parameters), are required to observe voice effects.

An interesting finding that, at first pass, might seem to contradict a word–voice binding hypothesis, emerges from many of these studies: item (e.g., word or sentence) and source (voice) memory can often be dissociated. Glisky et al. (1995) found a double dissociation among older adults, wherein better source memory (memory for the voice speaking a sentence) was correlated with higher frontal lobe function and better item memory (memory for the sentence itself) correlated with higher medial temporal lobe function. A set of follow-up experiments by Glisky et al. (2001) compared young and older adults for voice and sentence memory. Interestingly, the authors shifted item and source domains by having several different voices record two sentences (Experiments 1). Their purpose was to make voice the focal content (or "item") of interest, and the content of the sentence itself became the source component. This was done in an effort to discern whether frontal lobe function was specifically related to source memory or whether the frontal lobes become involved as a function of task difficulty. Evidence for the former hypothesis was found, and it was concluded that the frontal lobes are involved in processing of source information. A subsequent experiment (Experiment 4) in their report showed that source memory deficits associated with low frontal lobe functioning in older adults could be attenuated if attention was focused to the integration of sentence and voice at encoding. With respect to the young adults tested, there was a same-voice advantage in the item memory task (in this case, voice) and the authors suggested that young adults used a strategy that focused on the conjunction of voice and sentence features. In addition, a correlation between item and source memory for young adults – when given an integrative task at encoding – led to the suggestion that item and source memory may tap similar processes in young adults, at least when integration is highlighted at encoding. Requiring participants to attend not only to the source but to the source in relation to the item might be important to observing source memory effects. Recall that this notion was also suggested by Schacter and Church (1992). In 2008, Glisky and Kong expanded on this work and found evidence that integration of item and source requires less work in younger adults than in older adults, and that the frontal lobes are less involved in source memory for young adults. Taken together, it appears that the binding of item and source might be more automatic in young adults, and therefore the frontal lobes might be somewhat less implicated as neural correlates that underlie this binding in young adults (as opposed to older adults). Performance on item and source memory tasks, however, does not always coincide – in spite of binding – and we suggest that this can be accounted for by paradigm-specificity and attentional focus.

3. Voice effects in implicit memory

Several of the aforementioned studies looked at the effect of voice congruency on implicit memory as well as explicit memory tasks. Setting aside recognition memory, an important question when looking at voice congruency effects is whether voice is used differently or to a different extent by other memory systems. Implicit memory tests for words, for example, have been found to be more dependent on perceptual characteristics changing between study and test (see Craik et al., 1994). Voice information then, being perceptual, probably has a bigger effect on implicit memory than explicit. Questions that arise about voice congruency in implicit memory thus include: is voice congruency utilized in the same way as in recognition memory? Do potential congruency benefits last longer? Does partial congruency provide a greater facilitation than is seen in explicit memory?

In 1984, Jackson and Morton investigated the potential effect of voice congruency on auditory priming. Here, a between-group design was used, with a relatively heterogeneous subject pool that was not balanced between groups. Nevertheless, in the study phase, participants either heard words spoken in one of two voices or read them, while making a living/non-living judgment about each one. At test, all words were presented in white noise for identification. Due to the design, the time elapsed between study and test (for a given word) was between 5 and 30 min. Results indicated a priming effect for all old words compared to new words and that the priming effect was greater for words that had been heard at study compared with words that had been read initially. However, no difference was found between the same vs. different voice groups (i.e., no voice effects). It is important to note that a semantic judgment was used at encoding and that noise was only used at test in this study. Based on the TAP framework (Morris et al., 1977), it seems reasonable that voice effects might be observed in implicit memory if participants pay attention to perceptual (i.e., acoustic) features at study.

A few years later, Schacter and Church (1992) conducted a series of experiments looking at both implicit and explicit memory as a function of voice congruency. These experiments were meant to address a framework which states that implicit memory effects are the result of a pre-semantic perceptual representation system (PRS; see Schacter and Church, 1992). The idea was that superficial information about encoded items is stored in this PRS system, which drives priming. Results of these experiments indicated that voice congruency had no effect on priming when white noise was used in an identification task at test (independent of encoding task), but that an effect was observed when word stems were spoken clearly in a word-stem completion task at test. It was suggested that the white noise disrupted processing of the PRS in the right hemisphere. Additionally, when Schacter and Church did find voice reinstatement effects (i.e., using the auditory stem completion task) in their implicit test, the results were somewhat surprising. In one such experiment, participants performed either a pleasantness rating encoding task or a pitch rating encoding task; the same-voice advantage at test was significant following the pleasantness task but not the pitch task. Since the pleasantness task was meant to index semantic encoding, perceptual sensitivity at test seemed less likely in this case than following a perceptual encoding task (such as pitch rating). However, the authors suggest a possible confound in that participants had trouble dissociating pleasantness of the word from pleasantness of the voice. Further, the authors reason that a perceptual task could yield voice effects if voice attributes are stressed in relation to the word itself, at encoding. In their second experiment utilizing auditory word stem completion in the clear, Schacter and Church sought to more finely separate semantic and perceptual properties in the encoding tasks used. Participants performed a meaning rating task as the semantic encoding task and a clarity rating task (judging how clearly voices enunciated each word) as the perceptual encoding task. Results indicated that voice reinstatement improved performance on the implicit test following both encoding tasks, but performance was not greater following the clarity rating encoding task. In their discussion, the authors speculate that voice effects might not be affected by encoding task because encoding of voice is an obligatory component of speech perception (Goldinger et al., 1991; Schacter and Church, 1992).

In a subsequent study, Church and Schacter (1994) used the same clarity rating task in an implicit task that required identification of low-pass filtered words at test. Words were degraded at test using a low-pass filter (rather than being embedded in white noise) in an effort to determine whether voice effects on priming can be found in an auditory identification task. In fact, voice changes between study and test had an overall detrimental effect on the implicit memory task, with more priming occurring when the voice was the same at test as at study. Additionally, the experiments revealed that voice effects were larger for implicit memory tasks than for explicit word recognition memory tasks.

In his series of experiments, Goldinger (1996) also looked at implicit memory. In order to address the issue of white noise producing null results, Goldinger used words presented in noise both at study and at test (as opposed to words clearly studied that were tested in noise, as in Jackson and Morton, 1984; Schacter and Church, 1992). It was argued that the hypothesized increase in perceptual fluency between study and test should presumably bolster voice effects in implicit memory. In experiment 2, words presented in white noise were equally likely to be spoken in the same or a different voice between study and test, and participants always had up to 20 s to type the word presented (in both study and test phases). Results indicated that the same-voice advantage was prominent in the perceptual identification task for at least a week.

In a subsequent study, Sheffert (1998) directly explored the notion put forth by Goldinger (1996) — that voice effects can be found in noise when the same stimulus/mask combination is used at both study and test. Indeed, by looking at words in noise as well as words degraded by a filter, Sheffert found evidence for Goldinger's supposition. The finding of voice congruency effects using white noise clearly contradicts Schacter and Church's (1992) notion that the PRS system is disrupted by white noise. Sheffert (1996) had previously investigated implicit and explicit memory for words and voices in an attempt to elucidate whether voice and word information are stored together or separately. Not surprisingly, voice congruency between study and test resulted in increased priming on word identification tasks. This effect was bolstered when the type of processing engaged at study mirrored that engaged at test. Sheffert suggested that word and voice information are represented together in an episodic system, and that the primary determinant of voice change effects is the similarity of processing at study and at test (i.e., transfer-appropriate processing), irrespective of which type of memory the task primarily taps into.

In a study that assessed priming by the speed of response to old words, Meehan and Pilotti (1996) looked at the importance of voice familiarity, talker variability and voice congruency on such a measure. Importantly, the latter two aspects were investigated in a between-subjects design. The task used here was quite different than in the other studies described. Participants were asked to identify if the first phoneme of each word was "b", both at study and at test. Results indicated that increased talker variability (i.e., a greater number of voices used) had a detrimental effect on priming, which was partially mitigated by equating for familiarity with all voices. That is, if participants were given extra trials to familiarize themselves with the larger number of voices in the multi-talker scenarios, priming occurred in all but the most acoustically variable scenario. When familiarity was not taken into account, priming occurred when only one or two voices were used but not when three or four were employed. In terms of voice congruency, trials of same voice conditions were compared with trials of different gender voice conditions. Voice changes between study and test had a detrimental effect on priming, again indicating that voice information is essential to priming. The authors suggested that auditory priming was dependent on memory for sub-word components of stimuli. In sum, the findings support the idea that both talker variability and voice changes are potentially detrimental to priming when a phoneme monitoring task is used. It was suggested that this task oriented participants to sub-word processes, while a task such as word identification focuses on the whole word.

Pilotti et al. (2000) highlighted the importance of the type of task used, on whether voice effects are present or not. For our purposes, we limit discussion to the conditions referring to voice changes and auditory encoding only. In a direct comparison, Pilotti et al. investigated potential voice effects in four implicit memory tasks and two explicit memory tasks so that the only variable was task employed at test. It should be noted, however, that the experiment called for a between-subject design, so that participants only took part in one test. In all cases, participants performed a semantic encoding task while hearing words without background noise. Results indicated that voice changes between study and test impaired performance during auditory identification tasks (in noise and when low-pass filtered) and word recognition tasks, but not on completion tasks such as stem and fragment completion or cued recall. The results support Church and Schacter (1994) but contradict Schacter and Church (1992) and Jackson and Morton (1984). While it should be noted that Pilotti et al. have a more powerful experiment, informed by a greater number of trials, differences might also be the result of speakers used to record the stimuli. Specifically, Pilotti et al. used only two voices – one male and one female – as opposed to Schacter and Church, who used six different speakers. As such, it may be that voice information is less salient in the experiment by Pilotti et al., and/or that familiarity was probably a factor, given the few voices and the many trials that made up the experimental procedure. What is clear is that the effect of voice congruency on auditory implicit tasks is largely dependent on experimental design. When considering multi-talker babble as the type of noise employed in an auditory identification task, a problem occurs. Pichora-Fuller et al. (1995) suggested that the extra resources allocated to deciphering words in babble result in working memory deficiencies in such cases. Similarly, in her dissertation, Heinrich (2006) posited that encoding is limited in babble scenarios due to additional required processing of the words. Therefore, the paradigm used and especially the type of noise or degradation employed plays a crucial role in investigating voice effects in implicit memory.

4. Neural correlates of voice effects

In terms of imaging, a number of electrophysiological studies have investigated neural correlates of voice reinstatement in recognition memory. Wilding and Rugg (1996) used event-related potentials (ERPs) to investigate the role of source (i.e., voice) in recognition memory. Participants studied words in one of two voices, and the importance of voice was stressed in the study instructions. Participants were later tested visually, and results showed a typical parietal old/new effect wherein correctly recognized old words elicited a more positive deflection over parietal sites. When the source was also correctly attributed, the parietal old/new effect was larger. This parietal effect has been proposed to index recollection in a quantitative manner (e.g., Wilding et al., 1995). Wilding and Rugg (1996) also asked participants for source judgments, which referred to the voice in which words were originally presented at study. In addition to the parietal old/new effect, they also found a later right frontal positivity effect, posited to index context or source. The authors suggest that this "second [function], indexed by the right frontal effect, operates on the products of [the] retrieval process, and is necessary for the recovery of contextual information" (p. 903). Interestingly, this right frontal effect was still present when the study voice was incorrectly attributed, though it was much smaller in such cases. This study therefore represented recognition memory as a graded phenomenon. More accuracy appeared to correlate with greater positivity in the old/new waves.

Allan et al. (1998) suggested a dissociation between the left parietal and the right frontal effects. The first appears to be a task-independent retrieval effect and seems to be present regardless of the explicit memory task used. The left parietal old/new effect is described by Tendolkar et al. (2004) as a parietal positivity that onsets at 400 ms and lasts 400–600 ms, and has been shown to overlap the N2/P3 transition for auditory stimuli (Kayser et al., 2003). The right frontal post-retrieval (though see Senkfor and Van Petten, 1998) effect appears to be task-dependent, wherein the nature and amount of post-retrieval depends on the explicit memory task employed. Specifically, the right frontal effect indexes post-retrieval processing in recognition tasks (Wilding and Rugg, 1996; Wilding and Rugg, 1997). On the other hand, no effect is observed in associative recall tasks (Rugg, 1996) and a symmetrical deflection indexes post-retrieval processing in cued recall tasks (Allan et al., 1996). There is some indication that the post-retrieval effect is also prolonged for auditory stimuli (Kayser et al., 2003).

Senkfor and Van Petten (1998, p. 1005) attributed the right frontal effect to a "search for voice information." They separately tested memory for words and for speakers, specifically utilizing only old words in the source task (i.e., no old/new judgment required). Based on observed ERP latencies, including a source-related positivity over the right scalp region starting at 800 ms after word onset, they concluded that voice information was retrieved after word information, suggesting a hierarchical system. This contrasts with Pisoni's parallel system (Pisoni, 1993), which implies the possibility of simultaneous retrieval. However, it is important to note that the study by Senkfor and Van Petten emphasized semantic encoding. It might be that placing the emphasis on perceptual encoding could facilitate an earlier, perhaps more automatic retrieval of speaker in the source task. It should also be noted that automatic use of voice congruency could precede actual retrieval of voice information. For example, same-voice facilitation might improve memory without necessitating conscious awareness of the similarity; subsequent retrieval of voice characteristics might involve additional mechanisms (which might also explain why word and source recognition tests have sometimes yielded opposite results).

Not surprisingly, then, it has been suggested that attention plays a key role when considering the ERP modulation reflective of recollection. In their review of literature, Kuper et al. (2012) suggested that the parietal old/new effect is sensitive to perceptual changes (i.e., voice), and therefore might produce differences based on same vs. different voice congruency parameters, but only when such characteristics are attended/used in the task being employed. This implies that gradients reflective of voice congruency might be seen in the parietal old/new effect when participants are told to attend to voice at study.

Recent work in our laboratory has sought to extend our understanding of ERP correlates of memory when voice is varied between study and test. Campeanu et al. (2013) used a block design and told participants to attend to both word and voice at study. Findings showed a left parietal old/new gradient reflective of voice congruency in the word recognition test. Correctly recognized old words in the word recognition test corresponded to a more positive parietal deflection than correctly identified new words. Though visual inspection suggested a gradient based on voice congruency, in which correctly recognized old words spoken by the same speaker at study and at test produced the most positive wave, distinctions between voice congruency conditions were not statistically significant. Campeanu et al. also tested voice recognition and found that same speaker trials produced a significantly more positive modulation than different speaker trials, which peaked at about 700 ms over right frontal scalp region. This agrees with research that has shown that voice congruency effects are more pronounced when perceptual encoding is emphasized (Goldinger, 1996).

In a subsequent study, Campeanu et al. (2014) used a continuous recognition paradigm and did not instruct participants to attend to voice. Though a parietal old/new effect was found, no differences based on voice congruency were observed for this modulation. It was suggested that voice must be specifically attended in order to see a quantitative gradient in the parietal old/new effect. However, the benefit of reinstating the same voice between study and test did correlate with ERP negativities as early as 200 ms over frontal sites. Though such early activity over frontal sites has often been linked with familiarity (e.g., Curran, 2000), more recent research has suggested that this frontal modulation actually indexes conceptual priming (Voss and Paller, 2006, 2007; Paller et al., 2007). In their review, Voss and Paller (2008) noted a few investigations that had differentiated between measures of explicit memory and perceptual (Schott et al., 2002, 2005, 2006) or conceptual priming (Voss and Paller, 2006; Voss et al., 2007), stressing the importance of behavioral measures to accompany ERP findings in parsing out the underlying mechanisms at play. However, these investigations involved visual presentations of stimuli. An investigation comparing priming and explicit memory as it relates to voice in a purely auditory experiment is currently lacking.

5. Future directions

Based on this analysis of the past forty years of voice-related memory research, it appears clear that a voice congruency facilitation effect on word memory occurs only under specific conditions. That is, while specific voice reinstatement (i.e., same speaker) often appears to facilitate recognition memory even without attention to voice at study, a more general potential benefit of similar voices between study and test is less clear. In terms of explicit memory experiments utilizing unfamiliar voices, encoding methods appear to play a pivotal role. Voice congruency effects have been found when voice is specifically attended at study (i.e., when relatively shallow, perceptual encoding takes place). However, since perceptual encoding tasks often rely on judgments about voice, the question of whether any type of shallow encoding confers voice effects in recognition, or whether voice-specific encoding tasks are required, remains ambiguous. It could be that shallow encoding entreats voice congruency effects or it could be that voice effects are subserved by much more specific circumstances. Schacter and Church's (1992) and Glisky et al.'s (2001) findings suggest that experiments designed to assess voice effects in explicit memory using perceptual encoding tasks should focus on voice attributes in relation to the words being spoken. A future study might use such a task in a word recognition test with minimal distractions, and contrast it with a non-vocal perceptual encoding task, in order to differentiate between the possibilities.

Behavioral findings coincide with a parietal old/new recollection effect that distinguishes between correctly identified old words and correctly identified new words. This modulation reflects recollection and appears to be present for all correctly identified old words. However, though the presence of this modulation appears to be amodal, its amplitude has been found to be affected by source attributes such as voice. Specifically, we suggest that the parietal old/new effect reflects voice congruency only when voice is attended at study. Further research, with more trials conferring greater power, is required to confirm this speculation. For example, an experiment contrasting perceptual and semantic encoding tasks might be useful in determining the neural correlates of voice congruency effects. In a purely auditory experiment where voice congruency is varied between study and test, participants would be required to perform one of three encoding tasks: listing the number of syllables or the first letter of a word (shallow task that is not voice-specific), judging clarity of a spoken word (shallow task that is voice-specific) and making a living/non-living judgment for each word (semantic task). A word recognition test could then differentiate between the neural correlates associated with retrieval in each of the three groups. We propose that the parietal old/new effect would be present in all correctly recognized old words. We can also speculate as to the modulation of the effect across the three groups. A gradient reflective of voice congruency would likely be most fine-grained for the clarity judgment task. A voice reinstatement benefit might be present in the shallow task that is not voice-specific, but a detailed gradient might be less likely. Lastly, we predict that the semantic task would show no voice congruency modulations in the parietal old/new effect. Furthermore, since voice congruency effects on source recognition appear to also be paradigm-/task instruction-dependent, a source recognition task could be added to investigate how the late right frontal effect would be modulated as a function of voice congruency and encoding task.

With respect to implicit memory, voice congruency effects are mixed; the presence of voice effects appears to depend greatly on the task employed. Using a word identification task, perceptual similarity between study and test conditions confers voice congruency effects. When words are studied in quiet (i.e., no background noise) but tested for identification in noise, results concerning voice congruency effects have been mixed. In addition, the type of perceptual degradation employed appears to have a differential effect. When words were studied without background noise but tested for identification after being low-pass filtered, voice congruency effects have been found. While voice effects have sometimes been observed when white noise is used, the use of multi-talker babble does not confer the same results. To clarify some of these mixed findings, future research should employ a paradigm like the one used by Pilotti et al. (2000) to directly compare voice congruency effects across auditory implicit memory tests, with the alteration of using more than only two voices. As such, if sufficient power can be obtained, one might even be able to look at the effects of more fine-grained voice congruency effects on implicit memory. Presumably if implicit memory is more dependent on perceptual characteristics, a potential similarity gradient reflective of voice congruency (rather than just voice reinstatement) would be more prominent in implicit tests.

In terms of ERP modulations, a characterization of implicit memory with respect to voice congruency is currently lacking. As a suggestion, future ERP studies should employ an implicit memory task (found to be sensitive to voice congruency effects, see above) to investigate whether a voice similarity gradient between study and test would be reflected in ERP modulation(s), perhaps over frontal sites at early latencies (see Voss and Paller, 2008 for a review relating such a modulation to priming).

In this review, we have discussed the role of voice congruency on memory performance. Our aim was not to be exhaustive but rather to explore how speaker voice may facilitate verbal memory. There is considerable evidence supporting the notion that people remember words better when they are repeated in the same voice, but only under the right experimental conditions. When participants process the word–voice combination, the specific speaker reinstatement benefit on memory performance is robust and has been found to occur even when attention is not specifically focused on the voice. The resistance of this memory to interference from distractor tasks, for example, is unclear at this point (e.g., Naveh-Benjamin and Craik, 1995 vs. Goldinger, 1996). Regardless, the same speaker benefit in auditory verbal memory suggests that acoustic details (i.e., voice) might be encoded with the word into a Gestalt. Further research is needed to explore how the binding of acoustic and semantic information takes place.

Acknowledgments

This research was supported by grants from the Canadian Institutes of Health Research (MOP106619, C.A.), and the Natural Sciences and Engineering Research Council of Canada (NSERC) (C.A., F.C.).

References

Allan, K., Doyle, M.C., Rugg, M.D., 1996. An event-related potential study of word-stem cued recall. Cogn. Brain Res 4, 251–262.

Allan, K., Wilding, E.L., Rugg, M.D., 1998. Electrophysiological evidence for dissociable processes contributing to recollection. Acta Psychol. 98 (2–3), 231–252 (Special Issue: Fluency and Remembering).

Bradlow, A.R., Nygaard, L.C., Pisoni, D.B., 1999. Effects of talker, rate, and amplitude variation on recognition memory for spoken words. Percept. Psychophys. 61 (2), 206–219.

Campeanu, S., Craik, F.I.M., Alain, C., 2013. Voice congruency facilitates word recognition. PLoS One 8 (3), e58778.

Campeanu, S., Craik, F.I.M., Backer, K.C., Alain, C., 2014. Voice reinstatement modulates neural indices of continuous word recognition. Neuropsychologia 62, 233–244.

Church, B.A., Schacter, D.L., 1994. Perceptual specificity of auditory priming: implicit memory for voice intonation and fundamental frequency. J. Exp. Psychol. Learn. Mem. Cogn. 20 (3), 521–533.

Craik, F.I.M., Kirsner, K., 1974. The effect of speaker's voice on word recognition. Q. J. Exp. Psychol. 26 (2), 274–284.

Craik, F.I.M., Moscovitch, M., McDowd, J.M., 1994. Contributions of surface and conceptual information to performance on implicit and explicit memory tasks. J. Exp. Psychol. Learn. Mem. Cogn. 20 (4), 864–875.

Curran, T., 2000. Brain potentials of recollection and familiarity. Mem. Cogn. 28 (6), 923–938.

Dodson, C.S., 2007. Retrieval-based illusory recollections: why study–test contextual changes impair source memory. Mem. Cogn. 35 (6), 1211–1221.

Dodson, C.S., Shimamura, A.P., 2000. Differential effects of cue dependency on item and source memory. J. Exp. Psychol. Learn. Mem. Cogn. 26 (4), 1023–1044.

Dodson, C.S., Holland, P.W., Shimamura, A.P., 1998. On the recollection of specific- and partial-source information. J. Exp. Psychol. Learn. Mem. Cogn. 24 (5), 1121–1136.

Fisher, R.P., Cuervo, A., 1983. Memory for physical features of discourse as a function of their relevance. J. Exp. Psychol. Learn. Mem. Cogn. 9 (1), 130–138.

Geiselman, R.E., Bellezza, F.S., 1976. Long-term memory for speaker's voice and source location. Mem. Cogn. 4 (5), 483–489.

Geiselman, R.E., Bellezza, F.S., 1977. Incidental retention of speaker's voice. Mem. Cogn. 5 (6), 658–665.

Glisky, E.L., Kong, L.L., 2008. Do young and older adults rely on different processes in source memory tasks? A neuropsychological study. J. Exp. Psychol. Learn. Mem. Cogn. 34 (4), 809–822.

Glisky, E.L., Polster, M.R., Routhieaux, B.C., 1995. Double dissociation between item and source memory. Neuropsychology 9 (2), 229–235.

Glisky, E.L., Rubin, S.R., Davidson, P.S.R., 2001. Source memory in older adults: an encoding or retrieval problem? J. Exp. Psychol. Learn. Mem. Cogn. 27 (5), 1131–1146.

Goh, W.D., 2005. Talker variability and recognition memory: instance-specific and voice-specific effects. J. Exp. Psychol. Learn. Mem. Cogn. 31 (1), 40–53.

Goldinger, S.D., 1996. Words and voices: episodic traces in spoken word identification and recognition memory. J. Exp. Psychol. Learn. Mem. Cogn. 22 (5), 1166–1183.

Goldinger, S.D., Pisoni, D.B., Logan, J.S., 1991. On the nature of talker variability effects on recall of spoken word lists. J. Exp. Psychol. Learn. Mem. Cogn 17, 152–162.

Heinrich, A., 2006. Investigating the Influence of Babble on Short-term Memory Performance, (Doctoral Dissertation. Retrieved from ProQuest Dissertations and Theses, Database, AAINR07835).

Jackson, A., Morton, J., 1984. Facilitation of auditory word recognition. Mem. Cogn. 12 (6), 568–574.

Johnson, M.K., Nolde, S.F., De Leonardis, D.M., 1996. Emotional focus and source monitoring. J. Mem. Lang. 35, 135–156.

Johnsrude, I.S., Mackey, A., Hakyemez, H., Alexander, E., Trang, H.P., Carlyon, R.P., 2013. Swinging at a cocktail party: voice familiarity aids speech perception in the presence of a competing voice. Psychol. Sci. 24 (10), 1995–2004.

Kayser, J., Fong, R., Tenke, C.E., Bruder, G.E., 2003. Event-related brain potentials during auditory and visual word recognition memory tasks. Cogn. Brain Res. 16 (1), 11–25.

Kuper, K., Groh-Bordin, C., Zimmer, H.D., Ecker, U.K.H., 2012. Electrophysiological correlates of exemplar-specific processes in implicit and explicit memory. Cogn. Affect. Behav. Neurosci. 12, 52–64.

Meehan, E.F., Pilotti, M., 1996. Auditory priming in an implicit memory task that emphasizes surface processing. Psychon. Bull. Rev. 3 (4), 495–498.

Morris, C.D., Bransford, J.D., Franks, J.J., 1977. Levels of processing versus transfer appropriate processing. J. Verbal Learn. Verbal Behav. 16, 519–533.

Naveh-Benjamin, M., Craik, F.I.M., 1995. Memory for context and its use in item memory: comparisons of younger and older persons. Psychol. Aging 10 (2), 284–293.

Naveh-Benjamin, M., Craik, F.I.M., 1996. Effects of perceptual and conceptual processing on memory for words and voice: different patterns for young and old. Q. J. Exp. Psychol. 49A (3), 780–796.

Paller, K.A., Voss, J.L., Boehm, S.G., 2007. Validating neural correlates of familiarity. Trends Cogn. Sci. 11, 243–250.

Palmeri, T.J., Goldinger, S.D., Pisoni, D.B., 1993. Episodic encoding of voice attributes and recognition memory for spoken words. J. Exp. Psychol. Learn. Mem. Cogn. 19 (2), 309–328.

Pichora-Fuller, M.K., Schneider, B.A., Daneman, M., 1995. How young and old adults listen to and remember speech in noise. J. Acoust. Soc. Am. 97 (1), 593–608.

Pilotti, M., Beyer, T., 2002. Perceptual and lexical components of auditory repetition priming in young and older adults. Mem. Cogn. 30 (2), 226–236.

Pilotti, M., Bergman, E.T., Gallo, D.A., Sommers, M., Roediger, H.L., 2000. Direct comparison of auditory implicit memory tests. Psychon. Bull. Rev. 7 (2), 347–353.

Pisoni, D.B., 1993. Long-term memory in speech perception: some new findings on talker variability, speaking rate and perceptual learning. Speech Commun. 13, 109–125.

Roediger, H.L., McDermott, K.B., Pisoni, D.B., Gallo, D.A., 2004. Illusory recollection of voices. Memory 12 (5), 586–602.

Rosenblum, L.D., Miller, R.M., Sanchez, K., 2007. Lip-read me now, hear me better later: cross-modal transfer of talker-familiarity effects. Psychol. Sci. 18 (5), 392–396.

Rugg, M.D., Schloerscheidt, A.M., Doyle, M.C., Cox, C.J.C., Patching, G.R., 1996. Event-related potentials and the recollection of associative information. Cogn. Brain Res 4, 297–304.

Schacter, D.L., Church, B.A., 1992. Auditory priming: implicit and explicit memory for words and voices. J. Exp. Psychol. Learn. Mem. Cogn. 18 (5), 915–930.

Schott, B., Richardson-Klavehn, A., Heinze, H., Düzel, E., 2002. Perceptual priming versus explicit memory: dissociable neural correlates at encoding. J. Cogn. Neurosci. 14 (4), 578–592.

Schott, B., Henson, R.N., Richardson-Klavehn, A., Becker, C., Thoma, V., Heinze, H., et al., 2005. Redefining implicit and explicit memory: the functional neuroanatomy of priming, remembering, and control of retrieval. Proc. Natl. Acad. Sci. U. S. A. 102, 1257–1262.

Schott, B., Richardson-Klavehn, A., Henson, R.N., Becker, C., Heinze, H., Düzel, E., 2006. Neuroanatomical dissociation of encoding processes related to priming and explicit memory. J. Neurosci. 26 (3), 792–800.

Senkfor, A.J., Van Petten, C., 1998. Who said what? An event-related potential investigation of source and item memory. J. Exp. Psychol. Learn. Mem. Cogn. 24 (4), 1005–1025.
Sheffert, S.M., 1996. Implicit and explicit memory for words and voices. Diss. Abstr. Int. Sect. B Sci. Eng. 56 (11-B), 6431.
Sheffert, S.M., 1998. Voice-specificity effects on auditory word priming. Mem. Cogn. 26 (3), 591–598.
Tendolkar, I., Ruhrmann, S., Brockhaus, A., Donaldson, D., Wirtz, K., Fernández, G., et al., 2004. Neural correlates of temporal context discrimination. Biol. Psychol. 66 (3), 235–255.
Tulving, E., Thomson, D.M., 1973. Encoding specificity and retrieval processes in episodic memory. Psychol. Rev. 80 (5), 352–373.
Voss, J.L., Paller, K.A., 2006. Fluent conceptual processing and explicit memory for faces are electrophysiologically distinct. J. Neurosci. 26, 926–933.
Voss, J.L., Paller, K.A., 2007. Neural correlates of conceptual implicit memory and their contamination of putative neural correlates of explicit memory. Learn. Mem. 14, 259–267.
Voss, J.L., Paller, K.A., 2008. Brain substrates of implicit and explicit memory: the importance of concurrently acquired neural signals of both memory types. Neuropsychology 46 (13), 3021–3029.
Voss, J.L., Reber, P.J., Mesulam, M.M., Parrish, T.B., Paller, K.A., 2007. Familiarity and conceptual priming engage distinct cortical networks. Cereb. Cortex 18, 1712–1719.
Wilding, E.L., Rugg, M.D., 1996. An event-related potential study of recognition memory with and without retrieval of source. Brain J. Neurol. 119 (3), 889–905.
Wilding, E.L., Doyle, M.C., Rugg, M.D., 1995. Recognition memory with and without retrieval of context: an event-related potential study. Neuropsychology 33 (6), 743–767.
Wilding, E.L., Rugg, M.D., 1997. Event-related potentials and the recognition memory exclusion task. Neuropsychologia 35, 119–128.